

Statistical Analysis Report

1. Descriptive Statistics:

- **Mean Sales:** 123,650.48
- **Median Sales:** 97,955.5
- **Mode Sales:** 6,540
- **Standard Deviation:** 100,161.09

Interpretation: Sales are highly variable, with a right-skewed distribution. The mean is inflated by large transactions compared to the median.

2. Distribution Analysis:

- Histogram shows **right-skewed distribution**.
- **Shapiro-Wilk Test:** $p < 0.05 \rightarrow$ Data is **not normally distributed**.

3. Correlation Analysis:

- **Quantity vs Total_Sales:** 0.69 (Strong positive correlation)
- **Price vs Total_Sales:** 0.65 (Strong positive correlation)
- **Quantity vs Price:** 0.008 (No correlation)

Interpretation: Both price and quantity independently drive sales, with quantity being slightly stronger.

4. Hypothesis Testing:

- **T-test (East vs West):** $t = 2.02$, $p = 0.0496 \rightarrow$ Significant difference in mean sales.
- **ANOVA (Regions):** $F = 2.16$, $p = 0.097 \rightarrow$ No significant difference across all regions.
- **Two-way ANOVA (Region + Product):** $p > 0.13 \rightarrow$ Neither region nor product significantly affects sales.

5. Confidence Interval:

- **95% CI for Mean Sales:** (103,776.35 , 143,524.61)
Interpretation: True mean sales likely fall within this range, with $\sim \pm 20K$ margin of error.

6. Regression Analysis:

- **Model:** $\text{Total_Sales} \sim \text{Price} + \text{Quantity}$
- **$R^2 = 0.884$** $\rightarrow$ Explains 88.4% of variance in sales.
- **Intercept:** -121,700
- **Price Coefficient:** 4.61 $\rightarrow$ Each unit increase in price adds $\sim \$4.61$ to sales.
- **Quantity Coefficient:** 26,430 $\rightarrow$ Each additional unit sold adds $\sim \$26,430$ to sales.
- **Significance:** $p < 0.0001 \rightarrow$ Both predictors highly significant.

7. Business Insights:

- **Key Driver:** Quantity has a stronger impact on sales than price.
- **Recommendation:** Focus on strategies that increase units sold (bundling, promotions, regional targeting).
- **Regional Strategy:** While East vs West shows differences, overall regional averages are consistent — suggesting localized tactics may help but broad strategies should focus on product volume.